

Q17

Since the $10\ \Omega$ resistor and $30\ \Omega$ resistor are in parallel, they have the same p.d.

$$P_{10\Omega} = \frac{V^2}{10}, \quad P_{30\Omega} = \frac{V^2}{30}$$

Hence $P_{10\Omega} = 3 \times P_{30\Omega}$

$$\therefore \frac{P_{10\Omega}}{P_{total}} = \frac{3 \times P_{30\Omega}}{3 \times P_{30\Omega} + P_{30\Omega}} = \frac{3}{4} \text{ or } P_{10\Omega} = \frac{3}{4} \times P \text{ while } P_{30\Omega} = \frac{1}{4} P .$$

Ans: B

Q18